ABES ENGINEERING COLLEGE
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROBLEM STATEMENT

FaultGuard AI: Knowledge-Driven IoT Fault Diagnosis Assistant

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Academic Year: 2026–2027

1. Background

Industrial machines produce sensor readings such as vibration, temperature, current, voltage, pressure, and flow rate. These readings can help indicate abnormal operating conditions, but reviewing them manually and consistently can be difficult. If unusual patterns are not noticed promptly, maintenance teams may take longer to investigate a possible fault.

The project SRS identifies IoT data acquisition, MQTT-based communication, fault diagnosis, PostgreSQL storage, and dashboard visualization as the intended system scope. The current software implementation uses the SKAB Valve 1 dataset and a dataset-driven sensor simulator to demonstrate this workflow.

2. Problem Definition

The problem addressed by FaultGuard AI is the difficulty of monitoring machine-condition data and identifying potentially abnormal operating patterns from multiple sensor parameters in a timely, organized way. The project aims to provide a software prototype that receives sensor readings, applies a machine-learning anomaly detector, records diagnosis information, and presents sensor trends and status through a dashboard.

3. Intended Users

- Monitoring users who need to review sensor readings, trends, and fault status.
- System developers or maintainers who configure the data pipeline, diagnosis model, database, and dashboard.
- Academic evaluators who need to inspect the project architecture, implementation, and documented evaluation results.

4. Proposed Approach

1. Use SKAB Valve 1 dataset records as the source for a dataset-driven sensor simulation.
2. Publish sensor messages through MQTT to a configured HiveMQ broker.
3. Receive messages in an MQTT subscriber and apply a saved StandardScaler and Random Forest model to classify normal or anomalous patterns.
4. Apply rule-based checks to help assign a likely cause category and recommendation to abnormal readings.
5. Store received readings and diagnosis fields in PostgreSQL and write the latest record to a JSON file for dashboard use.
6. Display current values, status, and historical trends in a Streamlit dashboard.

5. Project Objectives

- Demonstrate a modular pipeline for sensor-data acquisition, communication, analysis, storage, and visualization.
- Use a trained machine-learning model to flag patterns that may be anomalous.

- Present sensor readings and diagnosis outputs in a dashboard that is understandable to monitoring users.
- Retain historical readings for reviewing trends and supporting further analysis.
- Document test status and model metrics transparently, distinguishing recorded artifacts from tests that have not been rerun or independently verified.

6. Success Criteria

- The source code provides the intended MQTT publisher/subscriber, inference, database, and dashboard components.
- When configured dependencies and services are available, valid sensor messages can be processed through the intended workflow and made available for display.
- The dashboard can present sensor values, anomaly status, and historical trends from an available data source.
- Model evaluation artifacts are documented with their source and limitations. The repository contains a recorded accuracy of approximately 93.17%; this is a project artifact and has not been independently reproduced for this document.
- Functional, integration, and deployment tests are recorded with actual outcomes before any claim of end-to-end success is made.

7. Scope and Limitations

In scope: dataset-driven sensor simulation, MQTT message flow, machine-learning-based anomaly detection, rule-based diagnostic hints, PostgreSQL persistence, and Streamlit visualization.

Out of scope: physical machine repair, direct control of industrial equipment, and hardware manufacturing. Physical ESP32 sensor deployment and end-to-end operation of the complete MQTT/database/dashboard pipeline have not been verified as part of this document. The project should therefore be presented as a software prototype using dataset-driven input unless a live hardware demonstration is separately completed.

8. Expected Outcome

The expected outcome is a prototype that demonstrates how machine sensor data can be organized into a monitoring and anomaly-detection workflow. It is intended to help users review abnormal patterns and possible diagnostic hints more conveniently. It is not a certified industrial safety system and should not be used as the sole basis for maintenance or safety-critical decisions.

9. Project Details

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Project Repository: https://github.com/aknsh78/CSE11-117-faultdiagnosis

Verification note: This document is based on the available project SRS and source-code descriptions. It does not certify that the complete system has been run successfully. Validate the MQTT broker connection, database writes, dashboard data path, and current model results before final submission.